

Problem Sheet 5

1.

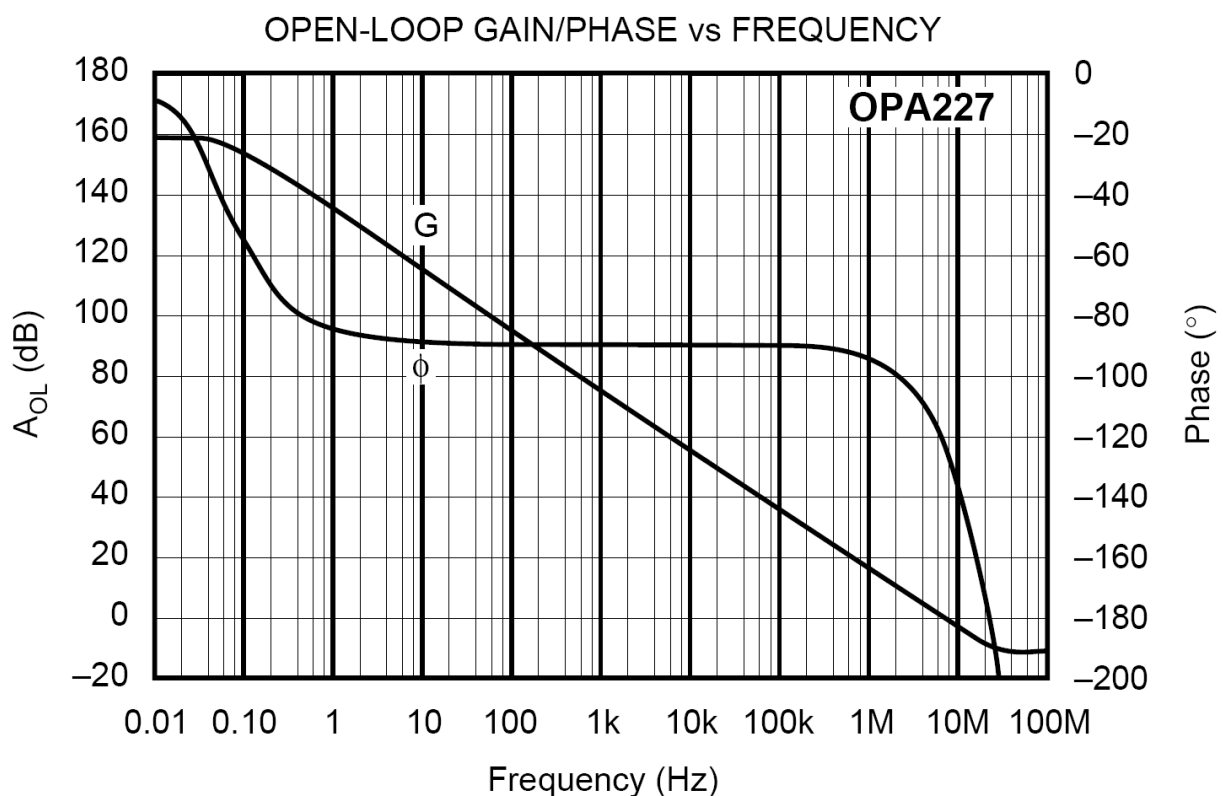
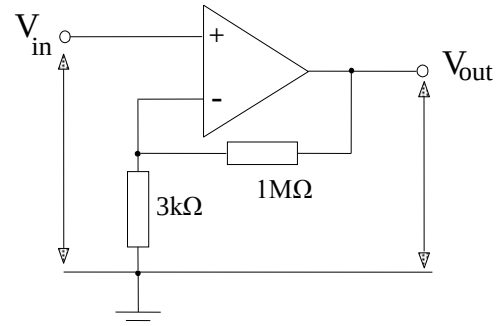
(a) A non-inverting amplifier is built according to the figure. The op-amp used is the OPA227 manufactured by Burr-Brown. The Bode plot for the open-loop amplifier is given in the data sheet and reproduced below.

Estimate the frequency at which the closed-loop gain curve will intersect the open-loop gain curve.

(b) Above this frequency, what is the closed-loop gain curve roll-off in dB/decade?

(c) Estimate the gain and phase margins. You may find it easiest to mark these on a copy of the figure.

(d) When connected closed-loop as a voltage-follower, will the amplifier be stable? Justify your answer.



2. SEMINAR QUESTION I

(a) Using a 'Flash' Analogue-to-Digital Converter (ADC), how many comparators are required to allow digitisation of a signal with a range 0 to 1 V with a digital resolution of 125 mV?

(b) How many digital bits will represent the measurement?

(c) Give the design of a Flash ADC which will meet the requirements of part (a).

(d) For each digital value, give the binary and decimal representations and the analogue input range.

Instrumentation

- (e) What will happen for an analogue input of +1.2 V, −0.125 V or +0.5 V?
- (f) What resistor values might you choose? Explain your answer.

3. SEMINAR QUESTION II

- (a) How many binary bits are required to represent all the digital numbers from 0 to 63 (decimal)?
- (b) The figure shown below is a weighted resistor Digital-to-Analogue Converter (DAC); the figure is taken from Horowitz & Hill. Here, each binary bit which is 'true' or '1' is represented by a voltage of 1 V. How many switches and resistors will be needed to accept the digital input range of part (a)?
- (c) Complete the diagram according to your answer for part (b). Note that this design uses 'real' resistances from the set of standard resistor values ([link](#)).
- (d) What is the output voltage for a digital input of 0, 1 or 63 (decimal)?
- (e) For the digital input numbers 2^n , where $n = 0 \dots 5$, sketch the output voltage versus input number. Is the device linear? Explain your answer.
- (f) What value of feedback resistor would be required to give an output exactly $-\frac{1}{10}$ times the input number? (Note you may ignore the 4.7k resistor, it is there simply to tie V_+ to ground potential such that both op-amp inputs see a similar impedance, and may be omitted in your design.)

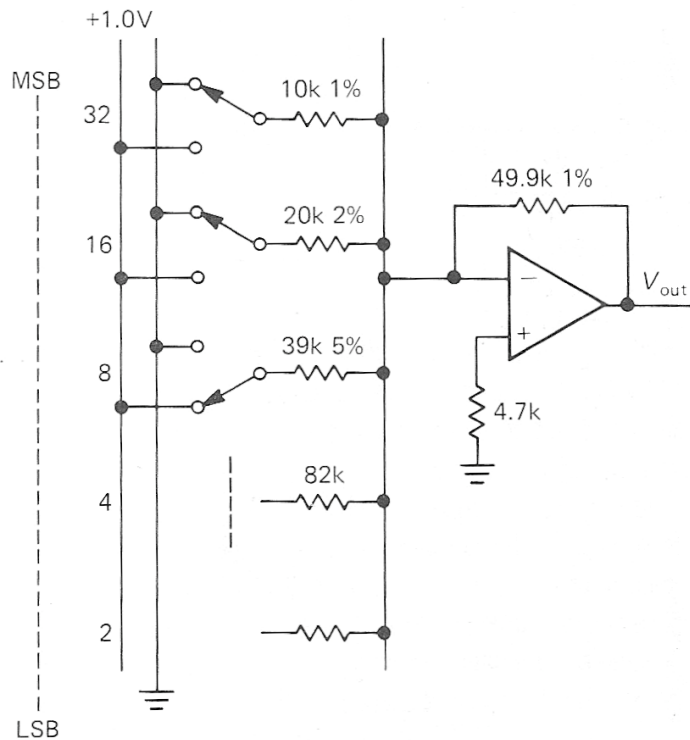


Figure 9.36

Numerical Answers and Hints

- 1 (a) about 20 kHz (b) -20 dB/decade (c) about 10 dB and 50°
- 2 (a) 7 (b) 3 (f) consider the function of the resistor chain – is much current needed?
- 3 (a) 6 (b) 6 (d) 0, -0.151V, -9.83V (e) Is it possible to match resistor values exactly? (f) Depends on choice of weighting resistors, however, 33k is a possible answer.